

DATABASE SYSTEMS LEARNING PROJECT

File Structures and Indexing

Hashing, B+ trees, and index organization

CONTRIBUTORS

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Question 1:

Part-a:

Question 1:

a-1

player.id

2981 $\rightarrow$ $0/05 = 1$

5252 $\rightarrow$ $0/05 = 2$

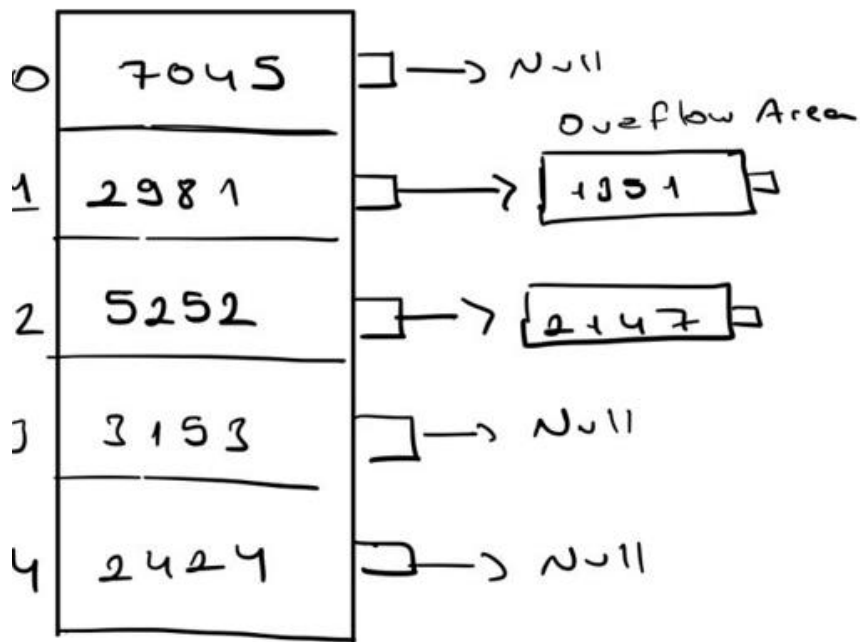
3153 $\rightarrow$ $0/05 = 3$

2424 $\rightarrow$ $0/05 = 4$

7045 $\rightarrow$ $0/05 = 0$

1351 $\rightarrow$ $0/05 = 1$

2147 $\rightarrow$ $0/05 = 2$



Part-b:

b-)

player_id + elo_score

$$2981 + 4023 = 4904 \text{ } 0/0.5 = 4 \checkmark$$

$$5252 + 959 = 6211 \text{ } 0/0.5 = 1 \checkmark$$

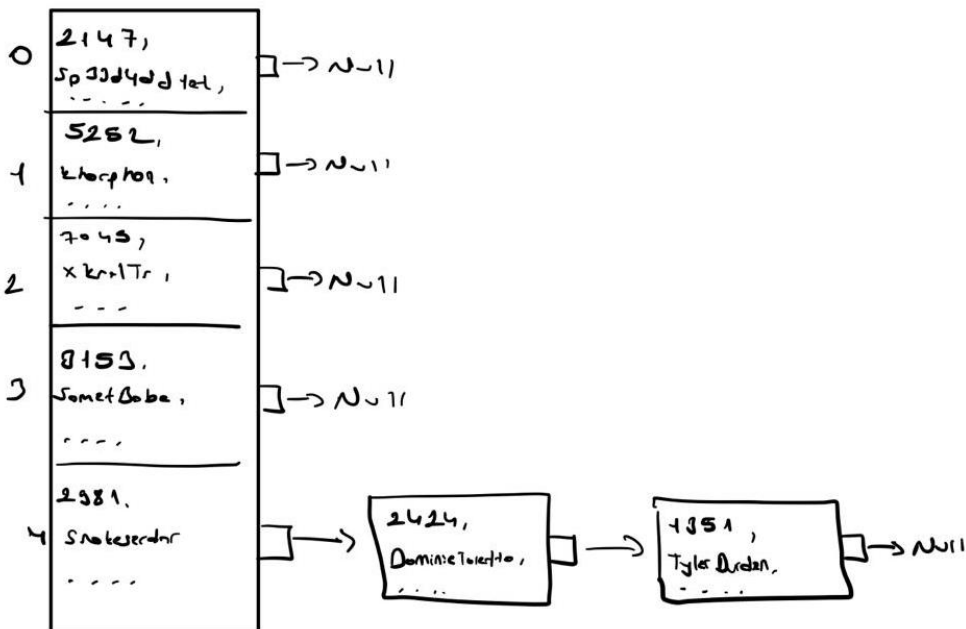
$$3153 + 1045 = 4198 \text{ } 0/0.5 = 3 \checkmark$$

$$2424 + 1520 = 3944 \text{ } 0/0.5 = 4 \checkmark$$

$$7049 + 1092 = 8141 \text{ } 0/0.5 = 2 \checkmark$$

$$1351 + 1173 = 2524 \text{ } 0/0.5 = 4 \checkmark$$

$$2147 + 1428 = 3575 \text{ } 0/0.5 = 0$$



Part-3:

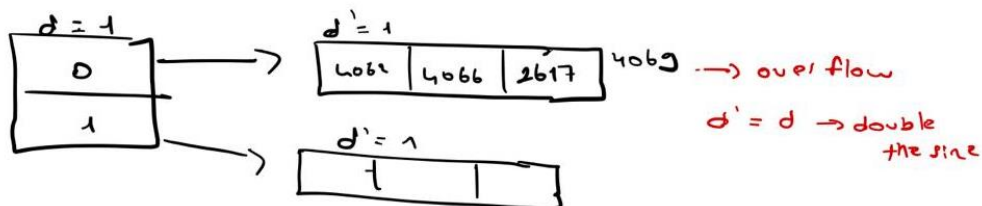
C-1 would choose first hash function because for the first table, records are distributed evenly, second table's distribution increased the overflow in certain buckets (bucket 4)

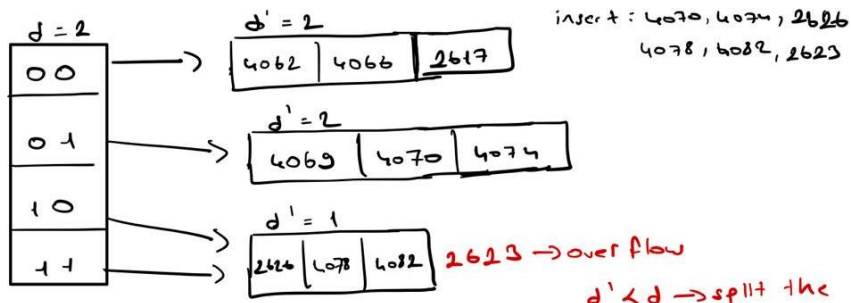
Question 2:

Question 2:

- ✓ 4062 % 29 → 2 → 00010
- ✓ 4066 % 29 → 6 → 00110
- ✓ 2617 % 29 → 7 → 00111
- ✓ 4069 % 29 → 9 → 01001
- ✓ 4070 % 29 → 10 → 01010
- ✓ 4074 % 29 → 14 → 01110
- ✓ 2626 % 29 → 16 → 10000
- ✓ 4078 % 29 → 18 → 10010
- ✓ 4072 % 29 → 22 → 10110
- ✓ 2633 % 29 → 23 → 10111
- ✓ 4086 % 29 → 26 → 11010
- ✓ 4088 % 29 → 28 → 11100
- 2610 % 29 → 0 → 00000

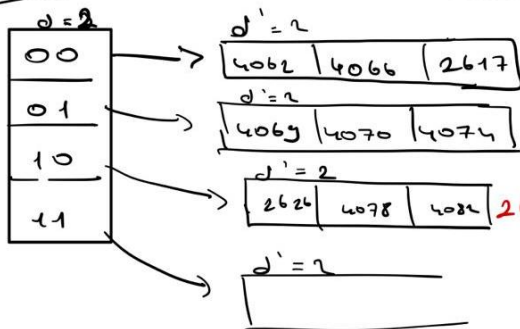
insert: 4062, 4066, 2617, 4069,



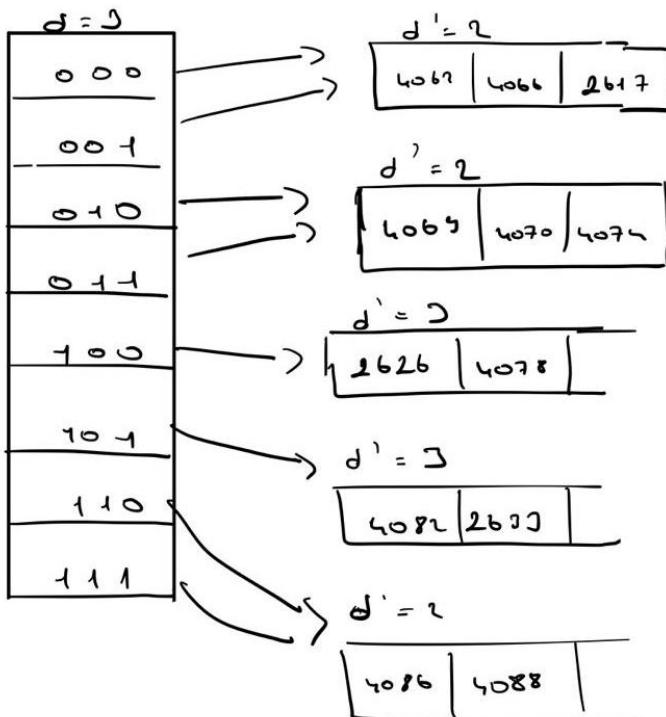


insert: 4070, 4074, 2626
4078, 4082, 2623

$d' < d$ → split the
bucket, no
need to double
the size



$d' = d$ so we
double the
size

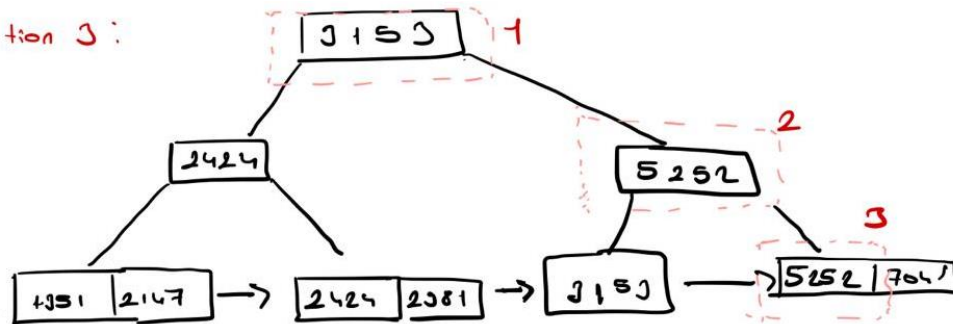


insert: 4086, 4088
2610

→ cause overflow
 $d' < d$ so split
and rearrange
records

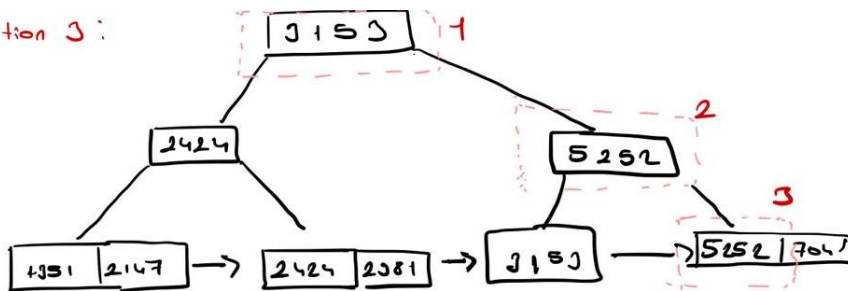
Question 3:

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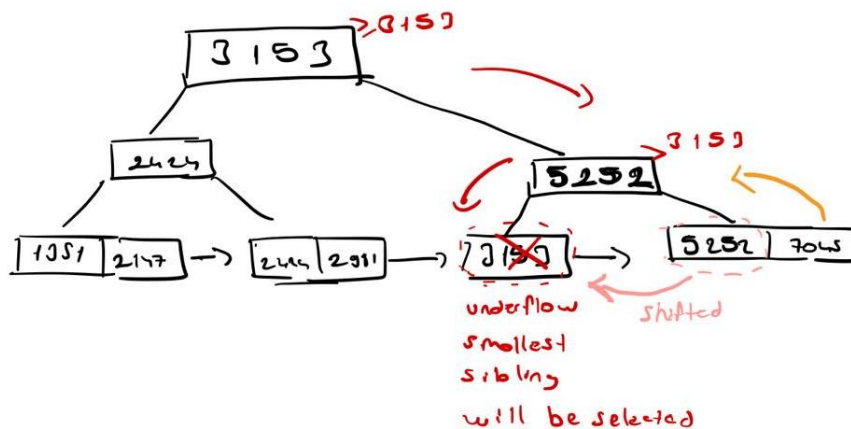
a-) Nodes 3153, 5252 and 5252 visited

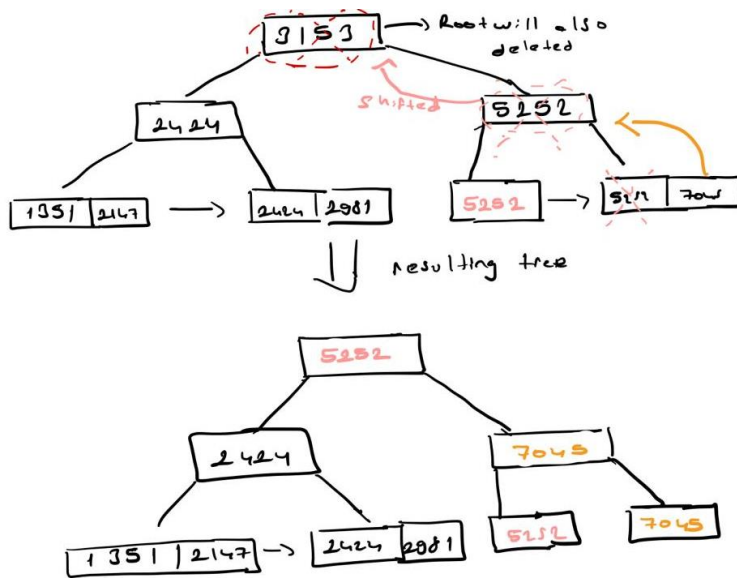
Question 3:



a-) Nodes 3153, 5252 and 5252 visited

b-) Deleting 3153





Question 4:

Part-a:

		race id	player id	car name	duration	position
Clustering field value	Block pointer	483	1351	doNotT4lk	4.57	2
1351		528	1351	SoapMachine	3.12	3
2147		848	1351	SoapMachine2	4.14	2
2424		432	2147	GottaGoF4st	5.13	8
2981		431	2424	FamilyFirst	4.12	2
3153		526	2424	FamilyFirstYou	0.55	1
3459		216	2981	lokмата1	5.31	5
		484	2981	lokмата2	4.56	1
		574	2981	lokмата3	4.32	1
		482	3153	DocHudson	5.01	3
		527	3459	McQueen	3.58	7

Part-b:

		race id	player id	car name	duration	position
Clustering field value	Block pointer	216	2981	lokмата1	5.31	5
DocHudson		431	2424	FamilyFirst	4.12	2
doNotT4lk		432	2147	GottaGoF4st	5.13	8
FamilyFirst		482	3153	DocHudson	5.01	3
FamilyFirstYou						
GottaGoF4st		483	1351	doNotT4lk	4.57	2
lokмата1		484	2981	lokмата2	4.56	1
lokмата2		526	2424	FamilyFirstYou	0.55	1
lokмата3		527	3459	McQueen	3.58	7
McQueen						
SoapMachine		528	1351	SoapMachine	3.12	3
SoapMachine2		574	2981	lokмата3	4.32	1
		848	1351	SoapMachine2	4.14	2